Identifying Key Regulators of NK Cell Dysfunction in Intrahepatic Cholangiocarcinoma Using Machine Learning

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***Abstract*—Intrahepatic cholangiocarcinoma (ICC) is a rare but aggressive primary liver cancer with limited treatment options and poor outcomes. A defining feature of ICC is its immunosuppressive tumor microenvironment (TME), where immune effector cells become functionally impaired. Although natural killer (NK) cells are present in high numbers, they often exhibit reduced cytotoxicity. The molecular mechanisms underlying this dysfunction remain obscure. In this study, we analyzed two ICC transcriptomic datasets from the Gene Expression Omnibus using CIBERSORTx to quantify resting and activated NK cell populations. We then applied machine learning approaches to identify genes that are associated with these NK cell functional states across both datasets. Genes were filtered through variance thresholding, ANOVA F-tests, recursive feature elimination, and elastic net regression, followed by repeated resampling to identify consistently enriched candidates. This work establishes an *in silico* approach for connecting gene-level expression patterns to immune cell dysfunction in solid tumors, and contributes to future studies aimed at restoring NK cell function in ICC.**

***Index Terms*—intrahepatic cholangiocarcinoma, natural killer cells, tumor microenvironment, immune evasion, CIBERSORT, machine learning**

I. Introduction

Intrahepatic cholangiocarcinoma (ICC) is a subtype of primary liver cancer that originates from the bile ducts in the liver, accounting for approximately 10% of all liver malignancies [1]. Although less common than hepatocellular carcinoma (HCC), which accounts for 75–85% of primary liver cancer [2], ICC remains a significant clinical challenge due to its late diagnosis and aggressive progression. Currently, the only curative treatment option for patients is surgical resection; however, the prognosis is extremely poor, with a five-year survival rate of below 10% [3]. The remaining treatment options are chemotherapy, radiation, and, most notably, emerging forms of immunotherapy.

Over the past decade, immunotherapy has shown potential across several cancer types such as renal cell carcinoma, neck squamous cell carcinoma, and lymphocytic leukemia through approaches including immune checkpoint inhibitors (ICIs), CAR-T cell therapy, and cancer vaccines [4], [5]. Unfortunately, in ICC, these therapies have limited effectiveness. This is due in large part to ICC's immunosuppressive tumor microenvironment (TME). The tumor microenvironment in intrahepatic cholangiocarcinoma is characterized by high infiltration of immunosuppressive cells like FOXP3$^+$ Tregs and tumor-associated macrophages (TAMs), along with low infiltration of cytotoxic CD8$^+$ T cells [6]. Collectively, these factors contribute to immune evasion and obstruct immunotherapeutic approaches.

Natural killer (NK) cells are innate cytotoxic lymphocytes capable of recognizing and eliminating malignant cells independently of antigen presentation. A previous study has reported that NK cells constitute a dominant immune population within the ICC tumor microenvironment, occupying the largest immune compartment relative to other infiltrating immune cells [7]. Despite this presence, NK cells have been observed to be functionally impaired or exhausted, suggesting suppression rather than absence [7]. This phenomenon, which is characterized by decreased cytokine production, reduced degranulation, and increased inhibitory markers, is well documented across various solid tumors and leukemias [8]. NK cells have immense therapeutic potential, especially given their success in treating metastatic tumors, which suggests that enhancing ICC NK cell activity may be a promising treatment route [9].

Bulk transcriptomic profiling provides a scalable means of investigating tumor-immune interactions, but directly linking gene expression patterns to immune cell function remains obscure. Computational deconvolution approaches such as CIBERSORTx enable the inference of immune cell subsets, including resting and activated NK cell populations, from bulk RNA sequencing and microarray data. Comparatively few studies have focused on distinguishing immune functional states or identifying features associated with immune activation and dysfunction.

In this study, we took a complementary computational approach to investigate the potential drivers of NK cell dysfunction in ICC. Traditional biological studies often focus on single targets, using methods like gene knockdowns, pathway inhibition, or protein staining to test hypotheses. Instead, we

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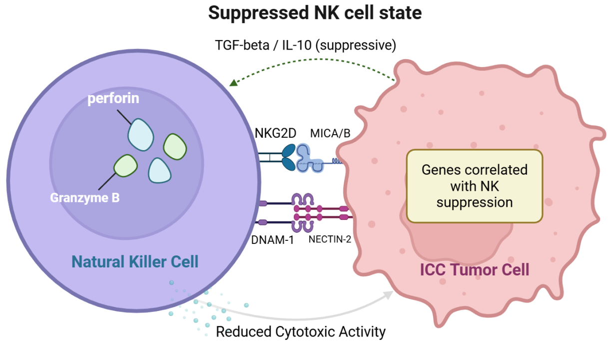


Fig. 1: Suppressed natural killer (NK) cell state in the intrahepatic cholangiocarcinoma (ICC) tumor microenvironment. Tumor-associated immunosuppressive signaling is associated with reduced NK cytotoxic activity, diminished granule release, and gene expression patterns correlated with NK suppression.

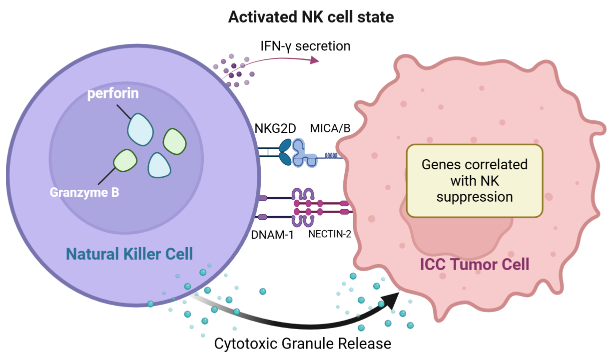


Fig. 2: Activated natural killer (NK) cell state in intrahepatic cholangiocarcinoma (ICC). Enhanced receptor–ligand interactions are associated with increased cytotoxic granule release and interferon-gamma (IFN-γ) secretion, alongside gene expression patterns correlated with NK activation.

analyzed transcriptomic datasets using CIBERSORTx to quantify resting and activated NK cell populations, then applied machine learning to identify genes consistently associated with these respective NK cell states. By emphasizing NK functional states, this work provides a hypothesis-generating approach for uncovering genes linked to immune suppression in ICC.

Our workflow combined several steps: variance thresholding to filter out low-variance genes, ANOVA F-tests to select top candidates, recursive feature elimination with logistic regression to narrow features further, and elastic net regression to produce a ranked list of genes. We repeated this pipeline across resampled subsets and retained only genes that appeared consistently.

The contributions of this work are threefold:

1) We created a reproducible pipeline combining statistical filtering with machine learning to identify regulators of NK cell functional states.
2) We identified a list of genes associated with NK cell suppression and activation across independent datasets.
3) We present an alternative to traditional biological methods, demonstrating how public transcriptomic data can generate testable gene lists for future validation.

The remainder of this paper is organized as follows. Section II reviews prior work on natural killer (NK) cell dysfunction in intrahepatic cholangiocarcinoma and related liver cancers, as well as computational approaches for analyzing tumor immune microenvironments. Section III describes the transcriptomic datasets, preprocessing steps, and immune cell deconvolution used in this study. Section IV details the feature scaling strategy and multistage feature selection pipeline used to derive NK cell–associated gene signatures. Section V presents the predictive modeling results, biological interpretation, and gene ranking analyses. Finally, Section VI discusses the implications of our findings, along with the limitations of the study and directions for future work.

II. Related Work

A. NK Cells in ICC and Other Liver Cancers

In intrahepatic cholangiocarcinoma (ICC), natural killer (NK) cell function has been sparsely documented. However, recent comprehensive reviews indicate that NK cells found within ICC tumors often exhibit reduced activity rather than strong cytotoxic function [10]. Single-cell transcriptomic analyses further suggest that while cytotoxicity-related signals can be detected, these likely reflected functionally exhausted NK populations. Studies have shown that adoptive NK cell transfer can suppress ICC tumor growth in nude mouse models [11], and that NK cytotoxicity is generally more potent than that of T cells against cholangiocarcinoma [12]. Furthermore, Porro and colleagues reported that several tumor-related factors may limit NK cell effectiveness in ICC, including poor infiltration into the tumor core and suppressive influences from the tumor microenvironment, contributing to overall NK dysfunction [10]. NK-dendritic cell crosstalk has also been shown to regulate liver cancer immune responses, with the study's authors concluding that NK-targeting therapies would positively affect the ICC TME [13].

B. Tumor Microenvironment in ICC

The tumor microenvironment (TME) of intrahepatic cholangiocarcinoma has been characterized as immunosuppressive, with a prominent amount of tumor-associated macrophages (TAMs), neutrophils, and regulatory T cells. Recent research has demonstrated that tumor-associated neutrophils (TANs) and TAMs interact to accelerate ICC progression by activating STAT3 signaling, therefore enhancing tumor invasion and metastasis [14]. Meanwhile, ICC-associated M2-polarized TAMs can drive tumor growth and invasiveness through an IL-10/STAT3-dependent epithelial-to-mesenchymal transition (EMT) [15]. Elevated TAM infiltration has been shown to correlate with worse prognosis [16].

Single-cell transcriptomic analyses of ICC tumors have identified distinct immunogenic cell death–associated subtypes with differing immune infiltration patterns, immune signaling activity, and macrophage-related transcriptional programs that correlate with patient prognosis [17]. Spatial multiomics profiling studies have further shown that the spatial organization of the ICC tumor microenvironment, contributes to disease progression, with immunosuppressive macrophage–T cell interactions associated with poorer survival [18]. Single-cell RNA sequencing studies have further demonstrated that the ICC tumor microenvironment is characterized by altered T-cell composition, including an increased ratio of regulatory T cells to central memory T cells, particularly in advanced disease stages, consistent with a progressively immunosuppressive immune landscape [19]. These analyses also identified tumor–immune communication pathways, such as SPP1–CD44 signaling between tumor cells and immune cells.

C. Computational Analyses of Tumor Microenvironments

To characterize immune landscapes, computational deconvolution methods have been developed to infer cell-type composition from bulk transcriptomic data. One widely used approach, CIBERSORT, uses support vector regression to estimate immune cell proportions from tumor RNA profiles [20]. Its extension, CIBERSORTx, enables single-cell–informed digital cytometry and cell type–specific expression reconstruction [21]. Other methods such as TIMER and UCSF xCell provide complementary statistical frameworks for immune infiltration analysis across The Cancer Genome Atlas (TCGA) and other cohorts [22], [23].

At the pancancer level, immunogenomic meta-analyses have leveraged these tools to define recurrent immune subtypes with prognostic and therapeutic relevance, demonstrating how conserved features of the TME shape clinical outcomes [24]. Analyses of bulk transcriptomic data have identified immune-related gene expression signatures linked to prognosis and patterns of immune infiltration in ICC [19]. Single-cell RNA sequencing further resolves heterogeneity within the TME, mapping dysfunctional NK and T cell states, macrophage polarization, and intercellular crosstalk in multiple solid tumors, including liver and breast cancers [25]–[27]. These computational frameworks provide strategies for analyzing ICC TMEs, which allows the modeling of bulk transcriptomic profiling.

While previous studies have characterized the immune cell composition in ICC and other liver cancers, most computational analyses focus on descriptive infiltration patterns or overall prognostic associations rather than links to specific immune cell functional states. In particular, bulk transcriptomic deconvolution studies typically examine total immune abundance or survival correlations, without distinguishing between resting and activated NK cell populations. This study integrates immune cell deconvolution with a feature selection pipeline to identify genes consistently associated with NK cell functional states across independent ICC datasets.

III. Methods

A. Pipeline Overview

We proposed a reproducible, multistage analysis pipeline to identify genes associated with natural killer (NK) cell activation inferred from bulk tumor transcriptomic data. As summarized in Figure 3, the pipeline consists of the following steps. (1) **Data Cleaning and Integration:** cross-platform normalization and batch correction were applied to combine microarray and RNA-seq expression profiles. (2) **NK Cell Estimation:** CIBERSORTx with the LM22 signature matrix was used to infer per sample proportions of resting and activated NK cells. (3) **Sample Labeling:** a binary NK label (NK-High vs. NK-Low) was assigned using a Bayesian Gaussian Mixture Model (BGMM) applied to the NK activation metric (NK_activated − NK_resting). (4) **Feature Scaling:** gene-wise Z-score normalization was applied. (5) **Data Splitting:** samples were randomly split into 80% training and 20% testing sets and evaluated over 40 independent trials. (6) **Four-Stage Feature Reduction:** variance thresholding, ANOVA F-statistic ranking, recursive feature elimination (RFE), and elastic-net regularized logistic regression were applied sequentially. (7) **Cross-Validation and Consensus Selection:** genes consistently selected across repeated trials were retained for downstream analysis.

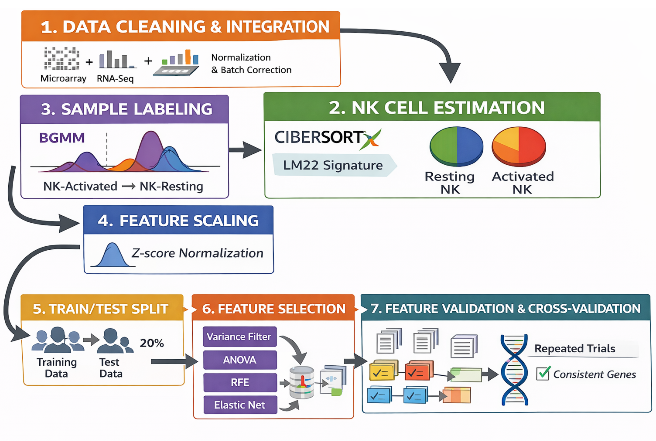


Fig. 3: Overview of the multistage analysis pipeline used to identify genes associated with NK cell activation. The workflow begins with data cleaning and cross-platform integration of microarray and RNA-seq datasets, followed by immune cell proportion estimation using CIBERSORTx. Samples are labeled as NK-High or NK-Low using a Bayesian Gaussian Mixture Model (BGMM) applied to the NK activation metric (NK_activated − NK_resting). Gene expression features are then Z-score normalized, split into training and testing sets, and subjected to a four-stage feature reduction pipeline consisting of variance thresholding, ANOVA F-statistic ranking, recursive feature elimination, and elastic-net regularized logistic regression. Final candidate genes are selected based on consistent appearance across repeated cross-validation trials.

B. Dataset Acquisition and Preprocessing

We obtained two publicly available transcriptomic datasets of intrahepatic cholangiocarcinoma (ICC) from the NCBI Gene Expression Omnibus (GEO) [28]. GEO is a public repository that stores raw and processed high-throughput functional genomic data. The datasets used were:

- **GSE107943** [29]: RNA-sequencing dataset comprising 30 tumor samples (raw counts available).
- **GSE32225** [30]: Microarray dataset comprising 149 tumor samples (raw CEL files / processed intensities available depending on GEO submission).

Only primary tumor samples were retained and adjacent normal or non-tumor samples reported in the metadata were excluded. Metadata includes sample IDs, platform annotation, and any removal notes. Because microarray and RNA-seq measure gene expression through different technologies and produce different numeric scales and distributions, we reprocessed each dataset through an appropriate, standardized pipeline and then combined them for analysis.

For the RNA-seq dataset GSE107943, expression data and sample annotations were retrieved using the `GEOquery` package in R [31]. Raw expression values were $\log_2$-transformed to stabilize variance and reduce the influence of extreme values. Genes with extremely low expression across samples were removed, as such features are unlikely to contribute meaningful signal and may introduce noise in high-dimensional modeling.

For the microarray dataset GSE32225, raw probe-level intensities were processed and normalized using platform-appropriate methods implemented in the `limma` package [32]. Probe identifiers were mapped to gene symbols, and probes mapping to the same gene were collapsed by summarization. To enable integration with RNA-seq data, the normalized microarray expression values were subsequently transformed using a cross-platform normalization procedure described by Foltz *et al.* [36]. This approach converts microarray measurements into expression units that are statistically comparable to RNA-seq data.

Following preprocessing, only genes common to both datasets were retained. This ensured that all further analyses were performed on a consistent gene set shared across platforms.

C. Estimation of Immune Cell Proportions

To quantify immune cell infiltration within the ICC tumor microenvironment, we applied CIBERSORTx to both harmonized datasets using the LM22 leukocyte signature matrix and 1,000 permutations. CIBERSORTx performs digital cytometry by matching bulk-sample expression against a reference signature matrix to estimate the relative fractions of 22 predefined leukocyte subtypes, including resting and activated NK cells. Although the LM22 reference was originally optimized for microarray platforms, the CIBERSORTx framework utilizes ν-Support Vector Regression (ν-SVR), which has been mathematically validated to provide high deconvolution accuracy across diverse platforms. This ν-SVR approach is inherently robust to outliers and cross-platform technical noise, ensuring reliable performance when applied to RNA-seq data.

CIBERSORTx reports *relative* cell-type proportions per sample: numeric values in the range $[0, 1]$ that indicate the fraction of the inferred immune compartment assigned to each cell type. An example of the extracted CIBERSORTx output used in this study is shown in Table I, illustrating the inferred proportions of resting and activated NK cells and the associated deconvolution confidence for representative tumor samples.

TABLE I: Example CIBERSORTx output showing inferred NK cell proportions for ICC tumor samples. Values represent unitless proportions of the inferred immune compartment per sample.

Sample ID	NK_resting	NK_activated
GSM0001	0.121	0.048
GSM0002	0.023	0.182
GSM0003	0.217	0.009
GSM0004	0.089	0.064

From each sample we extracted the `NK_resting` and `NK_activated` proportions. We computed an NK activation metric using the following equation for each sample and used it to derive a binary NK state (NK-High vs NK-Low) via Bayesian Gaussian Mixture Modeling (ref. Section III-D).

$$M_i = \text{NK\_activated}_i - \text{NK\_resting}_i \tag{1}$$

D. Definition of NK Functional States

CIBERSORTx deconvolution using the LM22 leukocyte signature matrix provides per-sample estimates for resting and activated natural killer (NK) cell proportions. To reduce these two related outputs to a single, interpretable quantity we compute the *NK activation metric*

$$M_i \;=\; \mathrm{NK}_{\text{activated},i} \;-\; \mathrm{NK}_{\text{resting},i}, \tag{2}$$

where $\mathrm{NK}_{\text{activated},i}$ and $\mathrm{NK}_{\text{resting},i}$ are the unitless CIBERSORTx proportions for sample i. Positive values $M_i > 0$ indicate a relative enrichment of activated NK signal; negative values indicate a relative enrichment of resting NK signal.

To derive biologically meaningful, reproducible binary labels (NK-high vs. NK-low) from the continuous metric M_i, we fit a two-component Bayesian Gaussian mixture model (BGMM). BGMM models the observed distribution of M as a weighted sum of Gaussian components:

$$p(M) \;=\; \sum_{k=1}^{2} \pi_k \, \mathcal{N}(M \mid \mu_k, \sigma_k^2), \tag{3}$$

where π_k are mixing weights ($\sum_k \pi_k = 1$), and $\mathcal{N}(\cdot \mid \mu_k, \sigma_k^2)$ is the normal density of component k. For each sample we compute posterior component membership probabilities

$$\gamma_{ik} \;=\; \Pr(z_i = k \mid M_i) \;=\; \frac{\pi_k \, \mathcal{N}(M_i \mid \mu_k, \sigma_k^2)}{\sum_{j=1}^{2} \pi_j \, \mathcal{N}(M_i \mid \mu_j, \sigma_j^2)}. \tag{4}$$

The fitted BGMM parameters for our cohort are:

$$\boldsymbol{\mu} = [-0.059268,\ 0.026267],$$
$$\boldsymbol{\sigma}^2 = [0.001374,\ 0.002807],$$
$$\boldsymbol{\pi} = [0.678713,\ 0.321287].$$

A labeling threshold between the two components is the mixture intersection $M^\star$, defined as the solution(s) to

$$\pi_1 \mathcal{N}(x \mid \mu_1, \sigma_1^2) \;=\; \pi_2 \mathcal{N}(x \mid \mu_2, \sigma_2^2). \qquad (5)$$

Here, $k \in \{1, 2\}$ indexes the latent mixture components, μ_k and σ_k^2 denote the component-specific mean and variance of the NK activation metric, π_k denotes the corresponding mixture weight, and z_i is the latent component assignment for sample i.

For the fitted parameters above, this intersection evaluates numerically to $M^\star = -0.000962$, which we plotted alongside the distribution in Figure 4. Because $M^\star$ lies very close to zero, we used the following labeling strategy for each sample in the main analysis

$$\text{label}_i \;=\; \begin{cases} 1 & \text{if } M_i \geq 0 \quad \text{(NK-high)} \\ 0 & \text{if } M_i < 0 \quad \text{(NK-low)} \end{cases}$$

as this aligns with biological interpretation and with the empirical BGMM separation.

Figure 4 shows the histogram of the NK activation metric with the two fitted Gaussian components, and the mixture intersection $M^\star = -0.000962$.

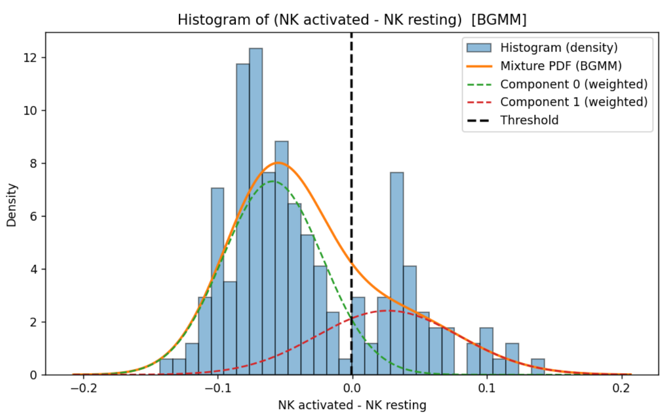


Fig. 4: Distribution of the NK activation metric $M_i = \mathrm{NK}_{\text{activated}} - \mathrm{NK}_{\text{resting}}$ across ICC samples with two-component BGMM overlay. The data-driven intersection $M^\star = -0.000962$ is represented as the dotted black line. Component parameters (means, variances, weights) are reported in the main text.

E. Feature Scaling

The combined dataset was split into training (80%) and testing (20%) partitions. Z-score standardization was performed in a gene-wise manner using statistics computed from the training set only. For each gene j, expression values were transformed according to:

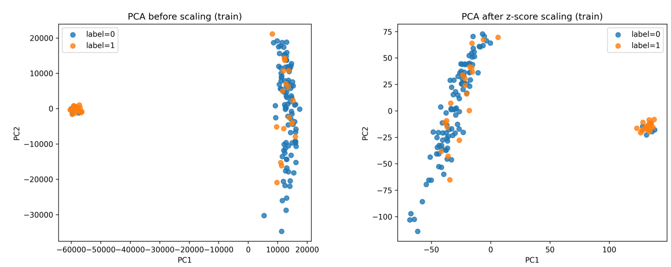
$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \qquad (6)$$

where x_{ij} denotes the expression of gene j in sample i, and μ_j and σ_j represent the mean and standard deviation of gene j calculated across training samples. The same parameters were subsequently applied to transform the test set, ensuring that no information from the test data leaked into the training process.

After standardization, each gene exhibited a mean of zero and unit variance within the training set. This scaling step facilitated stable optimization and improved comparability across features in high-dimensional transcriptomic space.

F. Principal Component Analysis

We used principal component analysis (PCA) to summarize global variance structure in the training set and to test whether the NK-high and NK-low groups occupy distinct regions of principal-component space. PCA was computed both on the raw (unscaled) expression matrix and on z-score standardized features (scaling parameters computed on the training partition only). Visual inspection and formal tests (described below) indicate that scaling reduces the dominance of a small number of very-high-variance features while preserving a statistically significant separation between NK-high and NK-low groups, as shown in Figure 5.



(a) PCA before scaling (training set).

(b) PCA after z-score scaling (training set).

Fig. 5: PC1 vs PC2 of the training set, before (Figure 5a) and after (Figure 5b) z-score standardization. Points are colored by binary NK label (0 = NK-low, 1 = NK-high). Scaling reduces the dominance of a few large-scale features (compare explained variance in Table II) while retaining statistically significant label separation along PC1.

Table II summarizes the numeric comparisons between raw and scaled PCA and group-separation tests. The fraction of variance explained by PC1 falls from 52.65% (raw) to 25.28% (scaled), indicating that a small number of high-variance features dominated the raw data. Despite this rebalancing, group separation remained statistically significant after scaling: two-sided Welch t-tests on PC1 gave $p = 1.51 \times 10^{-7}$ (raw) and $p = 4.55 \times 10^{-8}$ (scaled). Centroid-distance permutation testing (2,000 permutations) in PC1–PC2 space returned $p \approx 4.9975 \times 10^{-4}$ for both raw and scaled data. Silhouette scores by label were 0.4053 (raw) and 0.3397 (scaled), both indicating moderate separation. Cohen's d (PC1) effect sizes were large with $|d| \approx 1.589$ (raw) and $d \approx 1.629$ (scaled), indicating a strong group effect on PC1 even after scaling.

G. Feature Selection

Because transcriptomic matrices are extremely high-dimensional relative to our sample size (order of magnitude:

TABLE II: Summary of PCA metrics and group-separation tests (training set).

Metric	Raw (unscaled)	Scaled (z-score)
PC1 explained variance	0.5265	0.2528
PC2 explained variance	0.0894	0.0970
PC1 Welch p (t-test)	1.51×10^{-7}	4.55×10^{-8}
PC1 Mann–Whitney p	7.40×10^{-8}	1.74×10^{-8}
Centroid permutation p	4.99×10^{-4}	4.99×10^{-4}
Silhouette score	0.4053	0.3397
Cohen's d (PC1)	$\|1.59\|$	1.63

Notes. All statistics computed on the training partition. Centroid-distance permutation tests used 2,000 permutations.

$\sim$20,000 genes vs. $<$ 200 samples), we implemented a multistage feature-reduction pipeline (see Figure 3) to (1) remove uninformative and noisy genes, (2) select features with the strongest association to the binary NK label, and (3) identify genes with the largest predictive coefficients in a regularized model. The pipeline (variance thresholding $\rightarrow$ ANOVA F ranking $\rightarrow$ recursive feature elimination $\rightarrow$ elastic-net regression) was applied only to the training partition (80% of samples). Selected features were then used to transform the held-out test partition (20%) in each trial.

H. Variance Thresholding

Genes with near-constant expression across samples were uninformative for supervised models and added statistical noise. For each gene j, we computed sample variance on the training partition:

$$\mathrm{Var}(x_j) \;=\; \frac{1}{n_{\text{train}}-1}\sum_{i\in\text{train}}(x_{ij}-\bar{x}_j)^2. \tag{7}$$

We remove gene j if $\mathrm{Var}(x_j) < \tau$ with $\tau = 0.5$ (on z-scored data). On z-scored features, variance $= 1$, hence $\tau = 0.5$ removes very low-variance genes (std ≈ 0.707). This step is applied only to the training set.

I. ANOVA F-Test

After variance filtering we reduced the feature list to 1000 based on ranking from ANOVA F-Test, which detected features exhibiting the largest marginal difference between the two NK labels. The ANOVA F statistic for gene j (two groups: $g = 0, 1$) was calculated:

$$F_j \;=\; \frac{\mathrm{SSB}_j/(G-1)}{\mathrm{SSW}_j/(n-G)}, \tag{8}$$

with

$$\mathrm{SSB}_j = \sum_{g=0}^{1} n_g(\bar{x}_{j,g}-\bar{x}_j)^2, \qquad \mathrm{SSW}_j = \sum_{g=0}^{1}\sum_{i\in g}(x_{ij}-\bar{x}_{j,g})^2,$$

where $\bar{x}_{j,g}$ denotes the group mean for gene j, n_g the group size, and $G = 2$.

Genes with large F_j and correspondingly small p-values indicate strong marginal candidates; genes with $F_j \approx 0$ and $p \approx 1$ indicate no evidence of a mean difference and are deprioritized under a univariate criterion. Table III provides an example between three top-ranked and three bottom-ranked genes by F_j on the testing set, showing the magnitude difference between strongly and weakly discriminative features.

TABLE III: Illustrative ANOVA one-way results contrasting the top three and bottom three genes ranked by F-statistic

GENE	F-value	p-value
TMEM133	97.06215505	9.55765×10^{-18}
PIGK	96.95240710	9.87666×10^{-18}
UXS1	94.55544053	2.03108×10^{-17}
DCBLD2	2.8217×10^{-7}	0.999576917
RAP1GDS1	5.29727×10^{-8}	0.999816686
DERL3	1.23133×10^{-8}	0.999911619

J. Recursive Feature Elimination (RFE)

After univariate filtering with variance-thresholding and ANOVA F-Test(Section III-I), we applied a multivariate wrapper method — recursive feature elimination (RFE) — to select a compact set of genes whose combined expression best predicts the NK functional label.

While the ANOVA F-test ranks genes by marginal mean differences between classes, it does not account for correlations or redundancy among genes. In contrast, RFE evaluates each gene conditionally, measuring its contribution to the decision boundary given the presence of other genes. As a result, RFE retains genes that provide complementary information and removes features that are statistically significant in isolation but redundant in a multivariate model.

RFE uses a logistic-regression base estimator. For a standardized feature vector $\mathbf{z}_i$ the logistic model is

$$P(y_i = 1 \mid \mathbf{z}_i) \;=\; \sigma(\mathbf{w}^\top \mathbf{z}_i + b), \qquad \sigma(u) = \frac{1}{1+e^{-u}}, \tag{9}$$

with parameter vector $\mathbf{w} \in \mathbb{R}^d$ and intercept b. The model is trained by minimizing the regularized negative log-likelihood

$$\mathcal{L}(\mathbf{w}, b) = -\sum_{i=1}^{n}\Big[y_i \log\sigma(\mathbf{w}^\top\mathbf{z}_i + b) + (1-y_i)\log\big(1-\sigma(\mathbf{w}^\top\mathbf{z}_i+b)\big)\Big] + \lambda\|\mathbf{w}\|_2^2 \tag{10}$$

where λ is an ℓ_2 regularization parameter (the base estimator used inside RFE in our pipeline is a logistic regression with ℓ_2 penalty).

At each RFE iteration the fitted logistic model produces coefficient estimates $\widehat{\mathbf{w}} = (\widehat{w}_1, \ldots, \widehat{w}_d)$. We define the model-based importance score for gene j as the absolute coefficient magnitude:

$$r_j \;=\; |\widehat{w}_j|. \tag{11}$$

RFE removes the feature(s) with the smallest r_j, refits the estimator on the remaining features, and repeats this elimination/refit cycle until the target number of features k is reached.

RFE was configured to select a final subset of $k = 100$ genes from the ANOVA-filtered feature space. One feature was removed per iteration (step size = 1), resulting in a gradual elimination process in which coefficients were re-estimated after each removal. This conservative step size reduces instability in coefficient rankings that can arise when large groups of correlated features are removed simultaneously.

The base estimator used within RFE was a logistic regression model with an ℓ_2 penalty, optimized using the `liblinear` solver with a maximum of 10,000 iterations. The regularization strength was fixed at $C = 1.0$.

K. Elastic Net Logistic Regression

We trained a logistic regression classifier with an elastic-net penalty as the final classification and feature selection step. For standardized input $\mathbf{z}_i$:

$$P(y_i = 1 \mid \mathbf{z}_i) = \sigma(\mathbf{w}^\top \mathbf{z}_i + b), \qquad \sigma(u) = \frac{1}{1+e^{-u}}.$$

We minimized the elastic-net penalized negative log-likelihood:

$$\mathcal{L}(\mathbf{w}, b) = -\sum_{i=1}^{n} \Big[y_i \log \sigma(\mathbf{w}^\top \mathbf{z}_i + b) + (1-y_i) \log\Big(1 - \sigma(\mathbf{w}^\top \mathbf{z}_i + b)\Big) \Big] + \lambda \left(\alpha \|\mathbf{w}\|_1 + \tfrac{1-\alpha}{2} \|\mathbf{w}\|_2^2 \right). \tag{12}$$

where λ controls penalty strength and $\alpha \in [0, 1]$ trades off L1 vs L2 regularization. This is parameterized via inverse regularization strength C (so $\lambda \propto 1/C$) and $l1_ratio$ = α.

- The ℓ_1 **penalty** ($\|\mathbf{w}\|_1$) encourages *sparsity*: some coefficients are driven exactly to zero, which yields a compact set of predictor genes.
- The ℓ_2 **penalty** ($\|\mathbf{w}\|_2^2$) shrinks coefficients toward zero and stabilizes estimates when predictors (genes) are correlated. It reduces variance of the estimator without setting coefficients exactly to zero.
- **Elastic-net** combines these: it selects a small, interpretable set of genes (via ℓ_1) while retaining correlated groups more reliably (via ℓ_2).

Two elastic-net hyperparameters were tuned: the inverse regularization strength C and the mixing parameter α (`l1_ratio`). Hyperparameters were selected using inner cross-validation on the training partition only, optimizing mean validation area under the ROC curve (AUC). The search grid was:

$$C \in \{0.01, 0.1, 1, 10\}, \qquad \alpha \in \{0.1, 0.2, \ldots, 0.9\}.$$

For each candidate pair (C, α), a logistic regression model with elastic-net penalty was fit and evaluated, and the pair achieving the highest mean validation AUC was selected.

To obtain an unbiased estimate of predictive performance, we used a nested cross-validation strategy. In the outer loop, data were split into stratified training and validation folds. Within each outer training fold, inner cross-validation was used to select optimal hyperparameters (C, α). The model was then refit using these parameters and evaluated on the held-out validation fold. Performance was summarized using validation ROC AUC and classification accuracy. Across all 5 cross-validation folds, inner cross-validation consistently selected $C = 10$ and $\alpha = 0.1$.

IV. Results

A. Predictive modeling

Our final classifier was selected as the best-performing model among the methods evaluated (Table IV). Model selection prioritized overall discrimination and classification quality. The final model achieved an ROC AUC of 0.931 and an accuracy of 0.825. Precision, recall, and F_1 were each 0.85.

TABLE IV: Performance summary for the selected final classifier.

Metric	Final classifier
ROC AUC	0.931
Accuracy	0.825
Precision	0.85
Recall	0.85
F_1 score	0.85

As shown in Figure 6, the final classifier demonstrates strong discrimination between NK-low and NK-high samples. A curve that stays closer to the top-left indicates stronger separation between two labels.

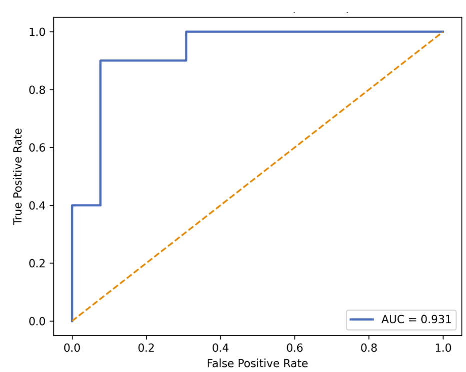


Fig. 6: Receiver operating characteristic (ROC) curve for the final PCA-based classifier built using genes retained by the elastic-net step.

B. Functional interpretation

To assess the biological relevance of genes associated with NK functional states, we performed pathway-level analysis on gene sets stratified by the sign of their average elastic net coefficients. Functional interpretation was carried out using the `clusterProfiler` package in R, with Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses. Statistical significance was evaluated using an adjusted p-value threshold of 0.05.

Genes negatively associated with NK activation showed significant enrichment in several biological processes, including regulation of translation, cytoplasmic translation, and miRNA processing. These enriched categories suggest increased involvement of post-transcriptional regulatory mechanisms in samples characterized by lower NK activity.

C. Gene ranking and interpretation

Elastic net–regularized logistic regression identified a set of genes consistently associated with inferred NK functional states across both ICC datasets. For each gene, the average model coefficient reflected the direction and magnitude of association with NK activity: positive coefficients indicated higher expression in samples classified as NK-high, while negative coefficients indicated higher expression in NK-low samples.

To assess robustness, we quantified the stability of each gene by tracking its selection frequency across repeated resampling and model refitting. The reported selection count corresponds to the number of iterations in which a gene was retained by the elastic net model, providing a measure of consistency across different training–testing splits.

Genes positively associated with activated NK cell abundance are listed in Table V, whereas genes negatively associated with NK activation are listed in Table VI. Genes exhibiting both high selection frequency and large absolute coefficients represented the most stable transcriptomic correlates of NK functional state, suggesting that their associations were reproducible and not driven by isolated samples or platform-specific effects.

Taken together, these results defined a transcriptional spectrum associated with NK activation and suppression in ICC. Rather than relying on individual genes in isolation, the ranked and stability-filtered gene sets provided a systematic framework for prioritizing candidate regulators of NK cell dysfunction for future experimental validation.

TABLE V: Top Genes Positively Correlated with Activated NK Cell Abundance

Gene	Count	Avg. Coefficient
SHC1	8	1.301
CSNK1E	8	1.249
ANKRD28	8	1.205
TMEM133	8	1.159
ACO2	8	1.002
IFIH1	7	1.336
HEATR3	7	1.315
STAMBP	6	1.249
AFTPH	6	1.056
PTBP1	5	0.951

V. Conclusions and Future Work

In this study, we developed a computational framework to investigate transcriptomic correlates of natural killer (NK) cell functional states in intrahepatic cholangiocarcinoma (ICC). By integrating immune cell deconvolution with machine learning, we identified gene signatures consistently associated with

TABLE VI: Top Genes Positively Correlated with Resting NK Cell Abundance

Gene	Count	Avg. Coefficient
NDNL2	8	-1.010
CNBP	7	-0.995
C5orf22	5	-1.090
ZC3H7A	5	-0.871
PPP2R5D	4	-0.973
CYP20A1	4	-0.886
ATG4A	2	-0.935
MKRN2	2	-1.120
GPBP1L1	2	-0.888
CASKIN2	3	-1.028

inferred NK activation and suppression across two independent datasets. This work emphasized NK functional state instead of immune cell abundance. Rather than establishing causality, this analysis was intended to be hypothesis-generating, providing a basis for prioritizing candidate regulators of NK dysfunction for future experimental validation.

A. Key gene associations with high NK activation

Our analysis identified a robust set of genes whose expression consistently correlated with NK cell states across independent datasets. The strong positive association of IFIH1 with activated NK cell abundance is a principal finding. *IFIH1* encodes the melanoma differentiation-associated protein 5 (MDA5). Previous studies have demonstrated that, in cancer models, the forced overexpression of MDA5 triggered tumor cell death and mobilized antitumor immunity, which included the activation of NK cells and T lymphocytes [34]. *IFIH1* initiates innate immune signaling to induce type I interferons (IFN-α/β). Type I IFN signaling is widely recognized as an upstream driver of NK-cell cytotoxicity, enhancing the expression of granzyme B and perforin, and facilitating NK-cell recognition and elimination of tumor cells [35]. This suggests that tumors with up-regulated IFIH1 may be crucial to increased immune surveillance.

SHC1 encodes an adaptor protein that links receptor tyrosine kinases to downstream oncogenic cascades. Although SHC1 is primarily known for promoting tumor growth and poor clinical outcomes, recent pan-cancer analyses have shown that its expression is also associated with enhanced immune cell infiltration and immune checkpoint activity across multiple tumor types [36]. In our model, SHC1 exhibited a strong positive correlation with activated NK cell abundance.

The gene CSNK1E encodes casein kinase 1 epsilon, which regulates numerous cellular processes, including circadian rhythm and DNA damage response. Elevated CSNK1E expression has been observed across multiple cancer types and is linked to tumor-cell survival and proliferation. A former study has shown that inhibition of CSNK1E induces cancer-cell-selective growth arrest and apoptosis [37]. However, CSNK1E has also been noted as a tumor suppressor in breast cancer [38]. Its positive correlation with activated NK cells may therefore reflect tumors in an immune-inflamed but signaling-active state, where both oncogenic and immune pathways are simultaneously unregulated.

B. Key gene associations with low NK activation

In addition to genes associated with activated NK states, our analysis identified several genes negatively correlated with activated NK cell abundance. One of the top genes with a high count was CNBP (cellular nucleic acid-binding protein) is a multifunctional protein that has been implicated in diverse tumor-intrinsic processes, with context-dependent roles in cancer progression. A previous study has shown that CNBP can promote tumor cell proliferation, invasion, and migration by acting as a transcriptional regulator of tumor-promoting genes, including matrix metalloproteinases and E2F2 [39]. Conversely, another study has demonstrated that CNBP overexpression can induce tumor cell death and suppress metastatic potential by downregulating heterogeneous ribonucleoprotein K [40]. In the setting of ICC, tumors characterized by elevated CNBP expression may therefore exhibit altered cellular programs associated with reduced NK activation.

ATG4A, or autophagy-related gene 4, has also been implicated in tumor immune evasion through its role in promoting cellular survival under stress conditions. ATG4A overexpression has been found in gastric cancer tissues, promoting cancer cell migration and invasion [41]. Similar roles for ATG4A have been reported in breast cancer, where elevated ATG4A expression correlates with poor clinical outcomes and resistance to endocrine therapy via enhanced autophagic flux [42]. Reviews of the ATG4 protein family also highlight its role in maintaining tumor cell survival under hostile microenvironmental conditions [43]. In this context, the negative association between ATG4A expression and NK activation may reflect tumor states that are less permissive to effective NK cell function. Tumors with elevated ATG4A expression may therefore favor stress-adapted, immune-evasive states characterizied by diminished NK activation.

C. Regulator Pathway Analysis

While deconvolution provides estimations of cell abundance, it cannot inherently delineate cell-type-specific expression. To address this, we mapped our top-ranked candidate regulators to established functional pathways using Reactome [44] and GeneCards [45]. This comparative analysis (Table VII) confirms that our identified regulators are documented mediators within signaling axes essential for NK-cell activation and tumor immune evasion, providing a biological basis for our predicted functional states.

D. Limitations and Future Directions

This study has several limitations. First, as bulk transcriptomic data conflates signals from multiple cell types and identifies associative rather than causal relationships, future validation using single-cell RNA sequencing, spatial transcriptomics, and *in-vivo* functional assays is necessary to confirm the roles of these regulators. Second, the rarity of Intrahepatic Cholangiocarcinoma (ICC) limits the availability of large-scale genomic datasets. However, by aligning our top-ranked features with established biological signaling axes, we prioritize mechanistic plausibility over statistical variance. Future work will focus on integrating these *in-silico* gene lists into targeted CRISPR-screening platforms to validate their potential as therapeutic targets for NK-cell restoration.targets for NK-cell restoration.

TABLE VII: Summary of Regulators and Associated Pathways

Gene	Pathway	Association with NK Functional State
IFIH1	IFN α/β (JAK-STAT)	Upstream component of Type I IFN induction; high expression aligns with the JAK-STAT signaling required for NK activation.
SHC1	IL-2/IL-15 Signaling	Adapter protein in the γc-cytokine axis; its presence supports the proliferative signaling environment of active NK cells.
ATG4A	Macroautophagy	Essential for autophagic flux; elevated levels are characteristic of tumor immune evasion and reduced NK-mediated lysis.

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